

Obed Junias

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RESEARCH AGENDA

I am **drawn to why questions**, like why a model behaves the way it does, why capabilities fail or hold up under certain conditions, and why interventions work or don't. A recurring thread in my work is **comparing how humans plan, reason, and act to how language models do the same, and using that gap to inform alignment**. Right now I am exploring this through mechanistic interpretability, structured reasoning, long horizon agentic planning, and alignment of both LLMs and agents, with two concrete projects: a mechanistic account of model collapse, and structured reasoning using entailment trees and search.

CURRENT INTERESTS

Mechanistic Interpretability, Model Collapse, Structured Reasoning, Long-Horizon Agentic Planning, LLM & Agent Safety and Alignment

EDUCATION

M.S. in Computer Science, University of Colorado Boulder Aug 2024 – May 2026
Courses: NLP, Deep Natural Language Understanding, Neurosymbolic NLP Methods (GPA: 4.0/4.0)
B.E. in Computer Science, BMS College of Engineering Aug 2017 – Aug 2021
Relevant Courses: Algorithms, Databases, Operating Systems, Machine Learning

PUBLICATIONS

- [1] **Junias, Obed**, and Pacheco, Maria Leonor. *LOGICAL-COMMONSENSEQA: A **Benchmark** for Logical Commonsense Reasoning*. **ACL 2026 (Main Conference)** [[short paper](#)]
- [2] **Junias, Obed***, et al. *Assessing Algorithmic Bias in Language-Based Depression Detection*. **IEEE-EMBS International Conference on Biomedical and Health Informatics (IEEE-BHI), 2025** [[paper](#)]

PREPRINTS & MANUSCRIPTS UNDER REVIEW

- [3] **Junias, Obed**, and Pacheco, Maria Leonor. *From Atomic Evidence to Logical Composition: Structured Compositional Reasoning over Compound Answer Options*. **Under review at EACL 2027** [[preprint](#)]

RESEARCH EXPERIENCE

Mechanistic Analysis of Synthetic-Data-Induced Model Collapse August 2026 - Present
Advisor: [Maria Pacheco](#), [BLAST Lab](#)

- Investigating **how model collapse manifests inside language model internals**, showing recursively-trained model generations are linearly decodable from activations with near-perfect accuracy.
- Comparing different training regimes to disentangle pure data drift from compounding parameter drift.
- Using **neuron ablation and activation patching** to causally localize where collapse-related information concentrates in the network.
- Testing whether **activation steering** can repair collapsed behavior, probing whether lost capabilities are gone or merely inaccessible.

Human-Centered AI Researcher August 2026 – Present
CU Boulder, NASA-funded project; Advisor: [Prof. Theodora Chaspari](#)

- Built an **NLP pipeline for participation and novelty metrics**, measuring speaking-time inequality, turn-taking, and informational redundancy.
- Designed visualizations that flag crew-level deviations in communication patterns to support team-functioning analysis in human-AI collaborative settings.

Structured Commonsense Reasoning and Benchmarking June 2025 – August 2026
Advisor: [Maria Leonor Pacheco](#), [BLAST Lab](#)

- Introduced **LOGICAL-COMMONSENSEQA** [1], a benchmark designed to isolate compositional and multi-fact reasoning failures in LLMs.
- Identified systematic mismatches between **chain-of-thought fluency and logical correctness**, demonstrat-

ing cases where coherent reasoning traces yield incorrect conclusions.

- Developed structured evaluation protocols to analyze reasoning depth and failure modes.
- Formulated answer selection as an ILP over atomic claims from compound options, using global constraints to reconcile inconsistent local predictions [3].

Responsible LLM Evaluation in Mental Health Applications

Jan 2025 – Nov 2025

Advisor: *Theodora Chaspari*, HUBBS Lab

- Built **large-scale LLM evaluation pipelines** to assess demographic bias and subgroup performance of LLMs (LLaMA, GPT-4) for mental health classification tasks.
- Studied how prompting strategies and fairness-aware losses affect reasoning consistency and robustness in low-resource and imbalanced data settings.
- Developed diagnostic metrics to characterize bias and reasoning instability beyond aggregate accuracy.
- Led research design for a **peer-reviewed publication** [3], demonstrating limitations of LLM reliability in sensitive domains.

Efficient Multi-Agent LLM Inference with Parameter-Efficient Adaptation

Aug 2025 – Dec 2025

- Designed a modular inference framework enabling scalable multi-agent LLM systems to support long-horizon, multi-step agentic workflows.
- Implemented dynamic **LoRA adapter loading and scheduling** under strict GPU memory constraints, streaming adapters from CPU to GPU at runtime.
- Conducted controlled experiments comparing adapter-based specialization against full-model switching, analyzing latency, throughput, and memory efficiency.
- Demonstrated that adapter-based agents reduce cumulative switching overhead by **20–90×**, making sustained multi-step agent workflows practical on constrained hardware.

Medical Ethics Assessment of Large Language Models

Jan 2025 – May 2025

- Designed evaluation protocols to assess ethical reasoning of LLMs in medical decision-making scenarios.
- Built a multiple-choice benchmarking pipeline to analyze how external knowledge affects reasoning consistency.
- Analyzed rationales and confidence signals to identify systematic failure modes in ethically sensitive settings.

INDUSTRY EXPERIENCE

Senior Member of Technical Staff at Oracle Corporation

Aug 2021 - Aug 2024

- Designed and implemented scalable automation and validation frameworks (**TestNG**, **BATS**) for end-to-end testing of **Oracle REST Data Services (ORDS)** and **SQLcl**, improving validation efficiency by **95%**.
- Led technical development, validation and delivery of a new Oracle Cloud **DB Tools** feature, coordinating a 15-engineer effort and achieving production readiness within 2 months.

Machine Learning Engineer Intern at Hewlett Packard Enterprise

Feb 2021 - Jul 2021

- Built OCR-based automation pipelines for processing unstructured enterprise documents, integrating learned document extraction into production workflows.
- Evaluated model performance across noisy and heterogeneous inputs, contributing to improvements in robustness and downstream automation reliability.

SKILLS

Languages & Frameworks	Python, PyTorch, Triton, LangGraph, SQL, Transformers, C++
LLM Systems	LLM Serving, LLM Evaluation, RAG Pipelines, Multi-Agent Systems and Tools
LLM Training	Pre-Training, Supervised Fine-Tuning, Transformers, PyTorch DDP
ML Infrastructure	CUDA, vLLM, FlashAttention, Quantization, GPU Memory Optimization

AWARDS AND RECOGNITION

- **NSF–EMBS–Google Young Professional NextGen Scholar**, *IEEE EMBS BHI Conference*, 2025